

Review

56%**9**

Out of 16 points

2:44

Avg Time Per Question

Your Answers:

Vignette

Jane Chang is an analyst at Alpha Fund covering the real estate and energy sectors. She and her colleague are analyzing two companies that are currently held by the fund.

The first company is Jupiter Corp., a publicly traded, national retail grocery store chain that has 2,800 physical stores. Jupiter leases most of its grocery stores and all five of its office locations that help the company achieve its core business of operating 50,000 square foot stores in all markets of the United States. Jupiter also owns the real estate (land and building) associated with 100 physical store locations. Jupiter recently announced that its board of directors approved a strategic real estate plan to pursue a separation of all its owned assets. The company currently has a speculative-grade credit rating.

The separation would be achieved through a series of sale-leaseback transactions with real estate investment trusts (REITs) that specialize in owning retail properties. Under the plan, Jupiter will sell its 100 owned grocery stores and lease them for 15-year terms with a combined annual rent expense of USD40 million. Jupiter expects to receive cash proceeds of approximately USD800 million from the property sales, which will be used to retire approximately USD600 million of debt and repurchase 4 million common shares.

Jupiter believes the pro forma capital structure following the transactions will enable it to receive an investment-grade credit rating. The sale-leaseback transactions value the 100 assets at an average capitalization rate of 5.50%. Based on Chang's colleague's research, the 25th, 50th, and 75th percentile cap rates for sale transactions for similarly situated properties and similar lease terms in the last three years were 5.00%, 5.50%, and 6.00%, respectively.

The second company is Saturn Corp., a publicly traded US energy company. Chang has been asked to assess the valuation of a potential spin off for this company. Saturn operates and reports three segments: Upstream, Midstream, and Downstream. In the last 12 months, the company reported the financial results shown in **Exhibit 1**.

Exhibit 1

Segment	EBITDA (USD millions)
Upstream	14,400
Midstream	5,760
Downstream	3,840
Consolidated	24,000

Saturn is currently trading at an enterprise value of USD408,000 million, or an EV/EBITDA multiple of 17. A spin off of the Downstream segment has long been rumored because it has been under-invested in by the current management team, resulting in slower revenue growth than its peers. Chang finds that the median Upstream, Midstream, and Downstream peers are trading at enterprise value-to-EBITDA multiples of 19, 17, and 13, respectively.

During an internal discussion, Chang's colleague makes the following three statements about the comparable company analysis method:

Statement 1: The method is not sensitive to market mispricing.

Statement 2: The estimates of value are derived directly from the market.

Statement 3: The method provides a reasonable approximation of a target company's value relative to similar transactions in the market.

1 Multiple Choice 0 / 1 point

Jupiter's strategic real estate plan would be best characterized as a:

☒ A. reorganization.

Correct Answer: **C. balance sheet restructuring.**

- ☐ B. cost restructuring.
- ☐ C. balance sheet restructuring.

Feedback

C is correct. Jupiter is undertaking a sale-leaseback transaction, which is a type of balance sheet restructuring. Jupiter would receive cash from the property sales and

recognize a liability equal to the present value of future lease payments. A is incorrect because a reorganization is a court-supervised restructuring process available in most jurisdictions for companies facing insolvency from burdensome debt levels and, sometimes, as a strategic measure to renegotiate contracts with unfavorable terms. B is incorrect because a cost restructuring refers to actions whose goal is to reduce costs by improving operational efficiency and profitability, often to bring margins to a historical level or to those of comparable industry peers.

2 Multiple Choice 1 / 1 point

Which of the following statements about Jupiter's motivations for the strategic real estate plan is incorrect?

- ☐ A. The transactions will enable Jupiter to sell a non-core business.
- ☐ B. The transactions will allow Jupiter to unlock the value of its real estate assets.
- ☒ C. The expected change in Jupiter's credit rating after the transactions will increase the firm's costs of capital.

Feedback

C is correct. The sale leasebacks will improve Jupiter's balance sheet by retiring debt and likely improving its credit rating, which will decrease (not increase) its costs of capital. A is incorrect because real estate ownership represents a non-core business for Jupiter. Its core business is operating 50,000 square foot stores in all markets of the United States. B is incorrect because the sale leasebacks will allow Jupiter to unlock the value in its real estate assets. Jupiter would receive cash from the property sales and recognize a liability equal to the present value of future lease payments.

3 Multiple Choice 1 / 1 point

Which of the following statements *best* describes Jupiter's average capitalization rate for the sale-leaseback transactions? Jupiter's average capitalization rate:

- ☒ A. is supported by the comparable transactions.
- ☐ B. compares favorably to the comparable transactions.
- ☐ C. compares unfavorably to the comparable transactions.

Feedback

A is correct. Jupiter's average capitalization rate for the sale-leaseback transactions is 5.50%, which is supported by the median of the 25th, 50th, and 75th percentile cap rates for sale transactions for similarly situated properties with similar lease terms in the last three years (5.00%, 5.50%, and 6.00%). B and C are incorrect because Jupiter's average capitalization rate for the sale-leaseback transactions is 5.50%, which is supported by the median of the 25th, 50th, and 75th percentile cap rates for sale transactions for similarly situated properties with similar lease terms in the last three years (5.00%, 5.50%, and 6.00%).

4 Multiple Choice 1 / 1 point

Based on **Exhibit 1** and the peer median EV/EBITDA multiples, Saturn's estimated enterprise value is *closest* to:

- ☐ A. USD392,000 million.
- ☐ B. USD408,000 million.

☒ C. USD421,440 million.

Feedback

C is correct. USD421,440 million is calculated as follows:

Upstream: $\text{USD}14,400 \times 19 = \text{USD}273,600$.

Midstream: $\text{USD}5,760 \times 17 = \text{USD}97,920$.

Downstream: $\text{USD}3,840 \times 13 = \text{USD}49,920$.

Consolidated: USD421,440 million.

A is incorrect because USD392,000 million is incorrectly calculated by using the average of the peer segment multiples $[(19 + 17 + 13)/3 = 16.33]$ to estimate the consolidated enterprise value.

Upstream: $\text{USD}14,400 \times 16.33 = \text{USD}234,720$.

Midstream: $\text{USD}5,760 \times 16.33 = \text{USD}93,888$.

Downstream: $\text{USD}3,840 \times 16.33 = \text{USD}62,592$.

Consolidated: USD392,000 million.

B is incorrect because USD408,000 million is Saturn's current enterprise value, not the estimated enterprise value based on median peer multiples for all three segments.

5 Multiple Choice 0 / 1 point

Which of Chang's colleague's three statements is correct?

- ☐ A. Statement 1
- ☐ B. Statement 2
- ☒ C. Statement 3

Correct
Answer:

B. Statement 2

Feedback

B is correct. An advantage of the comparable company analysis method is that estimates of value are derived directly from the market. This approach is unlike the discounted cash flow method, in which the value is determined based on many assumptions and estimates. A is incorrect because the comparable company analysis method is sensitive to market mispricing. As an example, suppose that all the comparable companies are currently overvalued by the market. A valuation relative to those companies may suggest a value that is too high in the sense that values would be revised downward upon a correction. C is incorrect because the comparable company analysis method provides a reasonable approximation of a target company's value relative to similar companies (not transactions) in the market. It assumes that "like" assets should be valued on a similar basis in the market.

Vignette

Elaine Lee is an analyst at an investment bank covering the energy sector. She and her junior analyst are analyzing Stratton Oil Corporation.

Stratton Oil Corporation is a publicly traded, US-based energy company that just announced its acquisition of Midwest Oil Corporation, a smaller US-based energy company. Stratton will pay USD55 in cash and 2.25 Stratton shares for each Midwest share, for a total consideration of USD40 billion based on share prices just prior to the announcement. Stratton's current trading enterprise value just prior to the announcement was USD170 billion. Lee concludes that the acquisition does not signal a change in strategy or focus for Stratton.

Stratton expects to realize annual recurring cost synergies of USD350 million (pre-tax), primarily from efficiencies in oil exploration and production activities and savings in corporate costs. The achievement of the full USD350 million in synergies is expected by the end of the third year after the acquisition closes. Synergies are realized in the amounts of USD117 million, USD233 million, and USD350 million in Years 1–3, respectively, and cash costs of USD175 million, USD280 million, and USD395 million are incurred in Years 1–3, respectively.

Expectations for revenues and total operating expenses for Stratton and Midwest for the next three years prior to the acquisition announcement are shown in **Exhibit 1**.

Exhibit 1

Stratton and Midwest Year 1–3 Figures, Prior to Acquisition (USD millions)

Stratton	Year 1	Year 2	Year 3
Revenues	21,325	22,391	23,511
Operating expenses	16,525	17,351	18,219
Midwest			
Revenues	5,350	5,618	5,898
Operating expenses	3,050	3,203	3,363

Lee's junior analyst makes the following comment during a conversation with Lee:

The acquisition is considered immaterial in the initial evaluation step for Stratton because it does not signal a change in strategy or focus.

Stratton's offer valued Midwest at an enterprise value of USD40.6 billion, including USD4.3 billion of existing Midwest debt. To finance the consideration of USD55 in cash and 2.25 Stratton shares for each Midwest share, Stratton will issue 104 million new shares and raise approximately USD26 billion in new debt and fund the remainder with cash on hand. Following the close, Stratton expects its outstanding debt will be approximately USD62 billion. Prior to the acquisition, Stratton has 1.096 billion shares outstanding trading at USD125 per share. Lee wants to determine how much the weights of debt and equity in Stratton's capital structure will change assuming a constant share price and that the book value of debt equals its market value.

During an internal meeting, Lee asks if Stratton could have achieved its same goals by undertaking an equity investment or joint venture. In response, Lee's junior analyst makes the following three statements.

Statement 1: Acquisitions require substantially greater capital investments than equity investments.

Statement 2: Acquisitions and equity investments are similar in that they both allow the acquirer to gain control of the target.

Statement 3: Relative to joint ventures, equity investments provide more equal governance representation and require larger investments.

Lee conducted a sum-of-the-parts valuation of Stratton's three segments and calculated an estimated enterprise value of USD187 billion just prior to the announcement.

6 Multiple Choice 0 / 1 point

Based on **Exhibit 1**, the forecasted operating income in Year 3 for the combined Stratton and Midwest is *closest* to:

☐ A. USD7,432.

☐ B. USD7,782.

☒ C. USD8,177.

Correct Answer: **B. USD7,782.**

Feedback

B is correct. The forecasted operating income in Year 3 is calculated as follows:

Combined Stratton and Midwest Oil: Year 1–3 Figures (USD millions)

Revenues	Year 1	Year 2	Year 3
Stratton	21,325	22,391	23,511
Add: Midwest revenues	5,350	5,618	5,898
Combined revenues	26,675	28,009	29,409
Operating expenses			
Stratton	16,525	17,351	18,219
Add: Midwest	3,050	3,203	3,363
Subtract: synergies	(117)	(233)	(350)
Add: one-time costs	175	280	395

Revenues	Year 1	Year 2	Year 3
Combined operating expenses	19,633	20,601	21,627
Operating income (Rev-OpEx)	7,042	7,408	7,782

A is incorrect because it incorrectly excludes Year 3 synergies of USD350 million. C is incorrect because it incorrectly excludes Year 3 one-time costs of USD395 million.

7 Multiple Choice 1 / 1 point

Lee's junior analyst's comment about materiality is:

- ☐ A. correct.
- ☐ B. incorrect because the acquisition is considered a small acquisition.
- ☒ C. incorrect because the acquisition represents more than 10% of Stratton's enterprise value prior to the transaction.

Feedback

C is correct. Materiality can be defined along two dimensions: size and fit. Although the acquisition does not signal a change in strategy or focus for Stratton, the transaction is considered large and material because it exceeds 10% of Stratton's enterprise value prior to the transaction. The total consideration is USD40 billion, based on share prices just prior to the announcement; thus, it represents 23.5% of Stratton's enterprise value just prior to the announcement of USD170 billion. A is incorrect because materiality can be defined along two dimensions: size and fit. Although the acquisition does not signal a change in strategy or focus for Stratton, the transaction is considered large and material because it exceeds 10% of Stratton's enterprise value prior to the transaction. B is incorrect because the transaction is considered large and material because it exceeds 10% of Stratton's enterprise value prior to the transaction.

8 Multiple Choice 0 / 1 point

The weight of equity in Stratton's capital structure as a result of the acquisition of Midwest assuming Lee's two assumptions is closest to:

- ☒ A. 29%.

Correct
Answer:

B. 71%.

- ☐ B. 71%.
- ☐ C. 81%.

Feedback

B is correct. Following the close of the acquisition, Stratton expects its outstanding debt to total USD62 billion, after assuming USD4.3 billion in existing Midwest debt and issuing USD26 billion. Therefore, prior to the acquisition, Stratton had approximately $(62 - 4.3 - 26 =)$ USD31.7 billion in debt and $(1.096 \text{ billion shares outstanding} \times \text{USD125 per share} =)$ USD137.0 billion in equity, resulting in a mix of debt and equity of 19% and 81%, respectively. After the acquisition, Stratton will have USD62 billion in debt and $(1.096 \text{ billion} + 104 \text{ million} =)$ 1.2 billion shares outstanding, which, priced at USD125 per share, results in USD150 billion in equity, resulting in a mix of debt and equity of 29% and 71%, respectively. The change in capital structure is summarized below.

Stratton's Capital Structure before and after Acquisition of Midwest

Stratton Capital Structure	Pre-Acquisition	Post-Acquisition
Debt %	19%	29%
Equity %	81%	71%

A is incorrect because 29% represents the percentage of debt after acquisition. C is incorrect because 81% represents the percentage of equity prior to acquisition.

9 Multiple Choice 1 / 1 point

Which of Lee's junior analyst's three statements is correct?

- ☒ A. Statement 1
- ☐ B. Statement 2
- ☐ C. Statement 3

Feedback

A is correct. Acquisitions require substantially greater capital investments than equity investments. Acquisitions—not equity investments—allow the acquirer to gain control of the target. Relative to equity investments, joint ventures provide more equal governance representation and require larger investments. B is incorrect because acquisitions (not equity investments) allow the acquirer to gain control of the target. C is incorrect because relative to equity investments, joint ventures provide more equal governance representation and require larger investments.

10 Multiple Choice 1 / 1 point

Stratton's estimated conglomerate discount just prior to the announcement is:

- ☐ A. -USD17 billion.
- ☐ B. USD0.

☒ C. USD17 billion.

Feedback

C is correct. The estimated conglomerate discount just prior to the announcement is calculated as follows: Total estimated enterprise value from sum-of-the-parts valuation – Current trading enterprise value = USD187 billion – USD170 billion = USD17 billion. A is incorrect because it incorrectly subtracts the estimated enterprise value from the current trading enterprise value. B is incorrect because the estimated and current trading enterprise values are not equal.

11 Multiple Choice 1 / 1 point

A company seeking to increase vertical integration will *most likely* pursue a(n):

☒ acquisition.

Correct Answer Feedback: Correct because vertical integration is achieved through an acquisition. Moreover, a corporation may be dependent on another company for inputs or for distribution of its products. By acquiring that company, the acquirer will increase its vertical integration, which can result in lower costs and lower risks and provides a more compelling proposition to customers and investors.

☐ divestment.

Incorrect Answer Feedback: Incorrect because divesting assets typically would either reduce or have no effect on vertical integration. The source of performance improvement for the divested business may be increased management attention, focus, or effort and potential synergies with the acquirer.

☐ balance sheet restructuring.

Incorrect Answer Feedback: Incorrect because the motivation of a balance sheet restructuring is typically not vertical integration, but rather improving returns on capital. There are two general types of issuer-specific motivations for restructuring actions: opportunistic improvement and forced improvement. Opportunistic improvement includes actions that alter the business model, trim the cost structure, or modify the composition of the balance sheet—all with the intention to improve returns on capital. Forced improvements are actions taken to enhance returns on capital when profitability falls below investors' required rate of return.

12 Multiple Choice 0 / 1 point

An analyst gathers the following projections (in \$ millions) for the first year following Company B's acquisition by Company A:

	Company A	Company B
Revenues	\$15,000	\$4,500
Operating expenses	\$7,500	\$1,500

While these projections were made prior to the acquisition announcement, Company A expects revenue dis-synergies of \$250 million and cost synergies of \$300 million for the same period.

Based on this information, the projected operating income (in \$ millions) in the first year following the acquisition for the combined entity is:

☒ \$9,950.

Incorrect Answer Feedback: Incorrect because the dis-synergies are correctly included in the calculation but not the synergies.

(in \$ millions)	Company A	Company B	Combined
Revenues	\$15,000	\$4,500	\$19,500
Subtract revenue dis-synergies			– \$250
			\$19,250
Expenses	\$7,500	\$1,500	\$9,000
Cost synergies incorrectly added			+ \$300
			\$9,300
Operating income			\$19,250 – \$9,300 = \$9,950

☐ \$10,550.



Correct Answer Feedback: Correct because according to the calculations below, which show intermediate steps.

(in \$ millions)	Company A	Company B	Combined
Revenues	\$15,000	\$4,500	\$19,500
Subtract revenue dis-synergies			– \$250
			\$19,250
Expenses	\$7,500	\$1,500	\$9,000
Subtract cost synergies			– \$300
			\$8,700
Operating income			\$19,250 – \$8,700 = \$10,550

☐ \$11,050.

Incorrect Answer Feedback: Incorrect because the synergies are correctly included in the calculation but not the dis-synergies.

(in \$ millions)	Company A	Company B	Combined
Revenues	\$15,000	\$4,500	\$19,500
Revenue dis-synergies incorrectly added			+ \$250
			\$19,750
Expenses	\$7,500	\$1,500	\$9,000
Subtract cost synergies			– \$300
			\$8,700
Operating income			\$19,750 – \$8,700 = \$11,050

13 Multiple Choice 0 / 1 point

An analyst gathers the following information in conducting comparable transaction analysis of an acquisition. For her analysis, she uses the median of valuation ratios from three recently closed comparable acquisitions.

Valuation Variable	Acquired Company 1	Acquired Company 2	Acquired Company 3
Acquisition share price	\$30	\$40	\$50
EPS	\$2.50	\$2.20	\$3.00

If the target company had an EPS of \$2.50 and was acquired at a price of \$40 per share, the analyst should conclude that the acquirer:

☐ underpaid for this transaction.

Correct Answer Feedback: Correct because the P/E ratio for the three companies is $\$30/\$2.50 = 12$ for Company 1, $\$40/\$2.20 = 18.2$ for Company 2 and $\$50/\$3.00 = 16.7$ for Company 3. The median of these three ratios is 16.7 and the estimated fair takeover value is $16.7 \times \$2.5 = \41.75 . Compared to an acquisition price of \$40, the acquirer underpaid for this transaction.

☐ paid fair value for this transaction.

Incorrect Answer Feedback: Incorrect because as demonstrated in the correct answer, the acquirer underpaid for this transaction.

☒ overpaid for this transaction.

Incorrect Answer Feedback: Incorrect because as demonstrated in the correct answer, the acquirer underpaid for this transaction.

14 Multiple Choice 0 / 1 point

With regard to an analyst's initial evaluation of a corporate restructuring, the size of a transaction affects:

☐ its timing only.

Incorrect Answer Feedback: Incorrect because materiality can be defined in this context along two dimensions: size and fit. The larger a restructuring, the more likely it is to affect an issuer's future cash flows and financial position and thus its value. It is true that the length of the timeline is largely determined by the size and complexity of the transaction. However, the size also affects the materiality of the transaction.

☒ its materiality only.

Incorrect Answer Feedback: Incorrect because an important consideration in the initial evaluation is timing. The length of the timeline is largely determined by the size and complexity of the transaction. It is true that materiality can be defined in this context along two dimensions: size and fit. However, the size also affects the timing of the transaction.

☐ both its timing and its materiality.

Correct Answer Feedback: Correct because materiality can be defined in this context along two dimensions: size and fit. The larger a restructuring, the more likely it is to affect an issuer's future cash flows and financial position and thus its value. Furthermore, an important consideration in the initial evaluation is timing. The length of the timeline is largely determined by the size and complexity of the transaction.

15 Multiple Choice 1 / 1 point

An analyst gathers the following information about a firm that is an acquisition target:

Most recent annual EBITDA (in millions)	\$356
EV (in millions)	\$6,000
EV/EBITDA range of peers	8x to 12x

If the acquirer expects to realize \$270 million in EBITDA synergies by Year 2, the acquisition multiple implied by this transaction:

- ☐ is lower than the range of peers.

Incorrect Answer Feedback: Incorrect because the multiple falls within the range of peers. One could arrive at this incorrect answer by subtracting the multiple before synergies from the multiple after synergies. In this case, $6,000/270 - 6,000/356 = 5.37x$, which is lower than the range of peers' multiples.

- ☒ falls within the range of peers.

Correct Answer Feedback: Correct because if synergies are included in the analysis, the acquisition EV/EBITDA multiple would be: $6,000/(270 + 356) = 9.58x$, which falls within the range of peers' multiples.

- ☐ is higher than the range of peers.

Incorrect Answer Feedback: Incorrect because the multiple falls within the range of peers. One could arrive at this incorrect answer if the synergies are not included in the analysis. In this case, the acquisition multiple would be: $6,000 / 356 = 16.85x$, which is higher than the range of peers' multiples.

16 Multiple Choice 1 / 1 point

A company has \$189 million in sales and \$17 million of EBIT. All else equal, if the company restructures its operations to achieve an EBIT margin of 20%, the required percentage reduction in its operating expenses is *closest* to:

- ☐ 9.9%.

Incorrect Answer Feedback: Incorrect because the percentage reduction in operating expenses is incorrectly calculated as EBIT divided by the operating expenses, or \$17 million / \$172 million = 9.9%.

- ☐ 11.0%.

Incorrect Answer Feedback: Incorrect because the percentage reduction in operating expenses is incorrectly calculated as the target EBIT margin minus the actual EBIT margin, or $20\% - 9\% = 11\%$, where the current EBIT margin is calculated as EBIT/sales, or \$17 million/\$189 million = 9.0%.

- ☒ 12.1%.

Correct Answer Feedback: Correct because the company's current EBIT margin is \$17 million/\$189 million = 9.0% which implies that the company had operating expenses of Sales – EBIT = \$189 million – \$17 million = \$172 million. To achieve an EBIT margin of 20%, EBIT needs to be \$189 million $\times$ 20% = \$37.8 million, and the company must decrease its operating expenses by \$37.8 million – \$17 million = \$20.8 million. In percent, the reduction is equal to \$20.8 million/\$172 million = 12.1%.